

GAYATRI KHARCHE

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EDUCATION

UNIVERSITY OF CALIFORNIA, SAN DIEGO

San Diego, CA

Master of Science in Machine Learning and Data Science

Sept 2023 - May 2025

Relevant Coursework: Statistical Learning, Data Science, Data Analysis, Linear Algebra, Machine Learning, Deep Learning, Visual Learning, Big Network Data, Random Processes, Recommender System, Data Mining, Statistical NLP

SAVITRIBAI PHULE PUNE UNIVERSITY

Pune, India

Bachelor of Engineering in Electronics and Telecommunication

Aug 2019 - May 2023

GPA: 4.00/4.00

SKILLS

Technical Skills: Python, SQL, MATLAB, C++, PyTorch, TensorFlow, Keras, Pandas, NumPy, Seaborn, Matplotlib, Scikit

Technologies: Data Science, Data Analysis, Data Visualisation, Machine Learning, Deep Learning, Cloud Computing, Predictive Modeling, Apache Spark, Tableau, AWS, Keras, LLM, Git, API, HTCondor, NLP, Docker, Kubernetes

WORK EXPERIENCE

UNIVERSITY OF CALIFORNIA SAN DIEGO

San Diego, CA

SDSC Student Researcher

Sept 2024 - Present

- Evaluating the Python binding of OSDF for use in AI workflows, including integration and optimization for Machine Learning and AI tasks.
- Conducting model ensembling to compare performance across various models, exploring methods to enhance AI and ML task efficiency, and submitting code in parallel to improve computational throughput.

SAN DIEGO SUPERCOMPUTER CENTER

San Diego, USA

Machine Learning Intern

July 2024 – Sept 2024

- Conducted performance benchmarking of machine learning models with data ranging from 150 MB to 150 GB using PyTorch on OSPool and utilized HTCondor to efficiently submit and manage computational jobs.
- Tested and benchmarked GPU performance on low-dimensional time series modeling tasks using CopulaCPTS.

HEWLETT PACKARD ENTERPRISE

Pune, India

Intern

Feb 2023 – Aug 2023

- Received intensive training in HPE GreenLake, hardware, network, cloud, server, VMware, and Linux fundamentals. Contributed to a VMware project showcasing expertise in domain establishment, virtual machine configuration, and implementation of DNS, NTP, and vCenter.
- Improved virtual environment performance, resulting in a 10% increase in operational efficiency. Proficient in using templates, content libraries, and migration strategies for streamlined virtualized environment management.
- Designed a scalable virtualized environment for testing and development, gaining hands-on experience in Windows Server Administration under Managed Services.

ORBITTAL HOUSE PVT. LTD.

Pune, India

Project Intern

Dec 2021 – Jan 2022

- Played a pivotal role in a Home Automation project, utilizing ESP8266, cloud technologies, and various platforms.
- Integrated data analytics features to optimize energy usage, enhance user experience, and provide insights.

UNIVERSITY PROJECTS

CUSTOM TRANSFORMER IMPLEMENTATION FOR NLP TASKS

October 2024 - November 2024

- Implemented a custom transformer encoder and GPT-like decoder from scratch for speech classification and language modeling, achieving strong classification accuracy and analyzing perplexity.
- Experimented with ALiBi positional encoding to optimize attention efficiency.

IMAGE INPAINTING WITH CONTEXT ENCODERS FOR CIFAR-10 DATASET

Jan 2024 - March 2024

- Implemented an auto-encoder for pixel context prediction, achieving a binary cross-entropy loss of ~ 0.54 .
- Trained Context Encoders for image inpainting, enhancing sharpness and demonstrating effective pre-training on CIFAR-10 with strong results.

NETWORK ANALYSIS OF INTERNATIONAL FOOTBALL MATCHES

April 2024-June 2024

- Analyzed 152 years of match data for 167 teams, constructing directed graphs using Python (NetworkX) and applying Louvain detection to uncover key team interactions and regional dynamics, with advanced visualizations in Gephi.